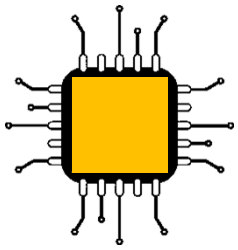
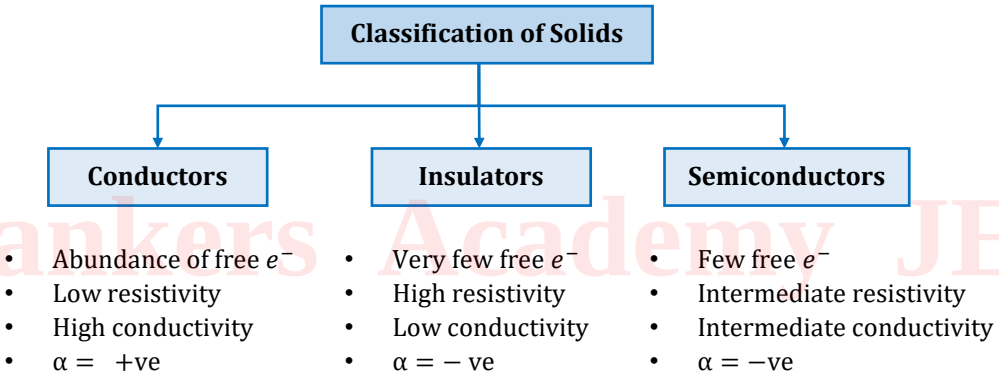
02 Semiconductors

We are surrounded by semiconductor devices and our day-to-day life depends on them.



Example: Mobile phone, Laptop, T.V., MRI Machine etc.

Material are majorly classified into three categories:



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Illustration 1

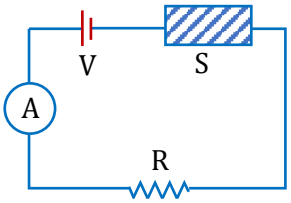
A Platinum rod and a Silicon rod are heated, what happens to their resistances?

Solution:

As temperature increases, resistance of platinum increases whereas that of silicon decreases.

Illustration 2

In the diagram 'S' is a semiconductor strip. Comment on the reading of the ammeter if temperature of strip is reduced.



Solution:

As temperature of strip decreases, resistance increases and hence current in circuit decreases so reading of ammeter decreases.

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Energy Bands in Solids:

Based on Pauli's exclusion principle:

In an isolated atom electron present in energy level that is according to Quantum Mechanical Laws, the energies of electrons in a free atom can not have arbitrary values but only some definite (quantized) values but in solid, atoms are not isolated, there is interaction among each other, due to this energy level split into different energy levels.

- This modification is not appreciable in the case of energy levels of electrons in the inner shells (completely filled).
- But in the outermost shells, modification is appreciable because the electrons are shared by many neighboring atoms.
- So, what used to be the discrete energy levels, now split-up or spread out to form energy bands.

Quantity of these different energy levels depends on the quantity of interacting atoms. Splitting of sharp and closely compact energy levels result into energy band. This is discrete in nature. Order of energy levels in a band is 10^{23} and their energy difference = 10^{-23} eV.

Energy band:

Range of energy possessed by electron in a solid is known as energy band.

Valence band (VB):

Range of energies possessed by valence electron is known as valence band.

(a) Have bonded electron.
(b) No flow of current due to such electron.
(c) Always fulfill by electron.
(d) This band is either partially or completely filled with electrons but never empty.

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Conduction band (CB):

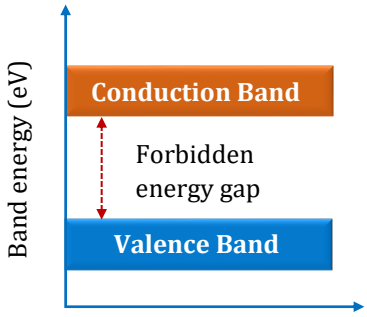
Range of energies/ possessed by free electron is known as conduction band.

(a) It has conducting electrons.
(b) Current flows due to such electrons.
(c) If conduction band is fully empty then current conduction is not possible.
(d) Electrons may exist or not in it.

Forbidden Energy Gap (FEG) (ΔEg):

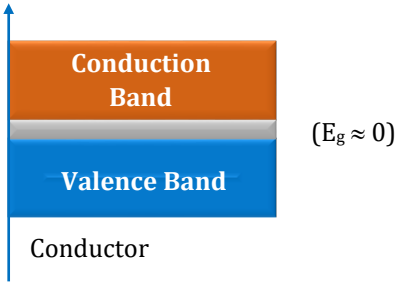
$$\Delta E_g = (C B)_{min} - (V B)_{max}$$

Energy gap between conduction band and valence band, where no free electron can exist.

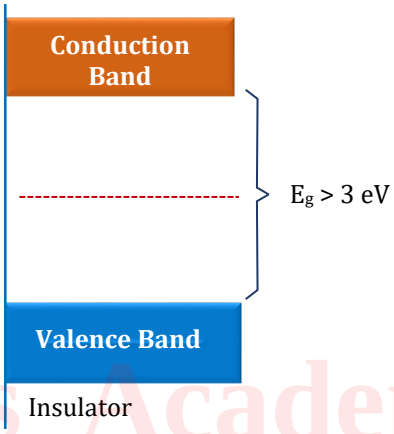


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- Width of forbidden energy gap depends upon the nature of substance.
- Width is more, then valence electrons are strongly attached with nucleus.
- Width of forbidden energy gap is represented in *eV*.
- As temperature increases forbidden energy gap decreases (very slightly).

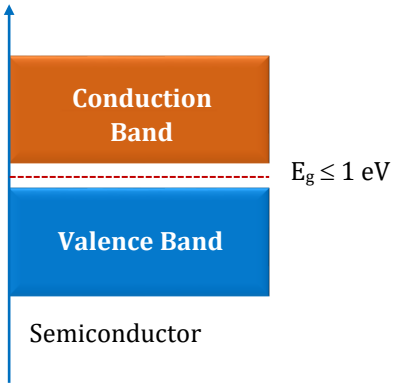


In conductors, there is no band gap since the valence band overlaps the conduction band.

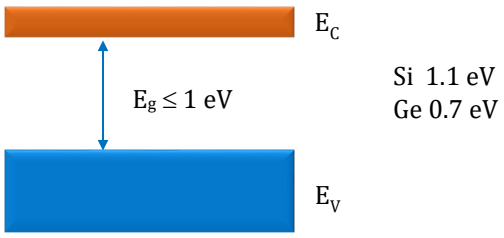


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The large energy gap between the valence and conduction bands in an insulator says that at ordinary temperatures, no electrons can reach the conduction band.



In semiconductors, the band gap is small enough that thermal energy can bridge the gap for a small fraction of the electrons.



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- At absolute zero temperature, no electron has energy to jump from valence band to conduction band and hence the crystal is an insulator.
- At room temperature, some valence electrons gain energy more than the energy gap and move to conduction band and can conduct under the influence of weak electric field.

Classification of Conductors, Insulators and Semiconductor:

On the basis of the relative values of electrical conductivity and energy bands the solids are broadly classified into three categories

(i) Conductor

(ii) Semiconductor

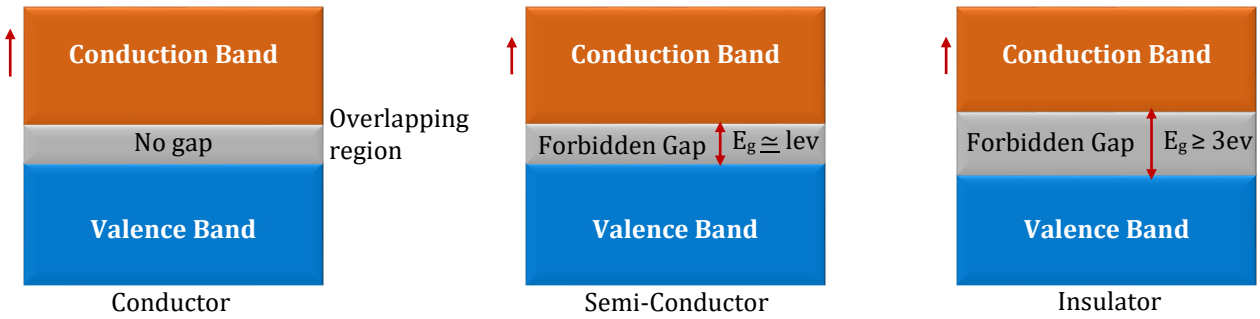
(iii) Insulator

Comparison between conductor, semiconductor and insulator:

Properties	Conductor	Semiconductor	Insulator
Resistivity	$10^{-2} - 10^{-8}\ \Omega m$	$10^{-5} - 10^{6}\ \Omega m$	$10^{11} - 10^{19}\ \Omega m$
Conductivity	$10^{2} - 10^{8}\ mho/m$	$10^{5} - 10^{-6}\ mho/m$	$10^{-11} - 10^{-19}\ mho/m$
Temp. Coefficient of resistance (α)	Positive	Negative	Negative
Current	Due to free electrons	Due to electrons and holes	No current
Forbidden energy gap	$\cong 0eV$	$\cong 1eV$	$\geq 3eV$

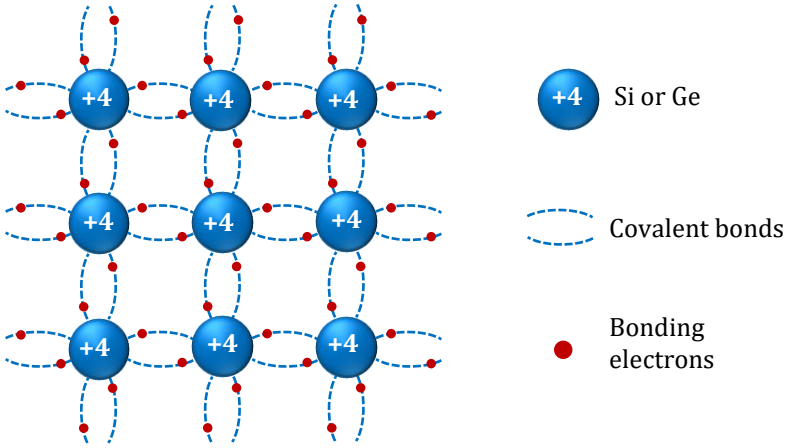
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Energy band diagram



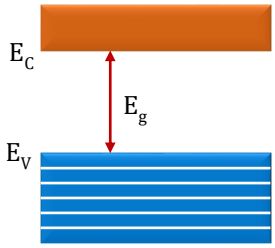
Example : Pt, Al, Cu, Ag GaF_2 Ge, Si, GaAs, Diamond, Wood, plastic, Mica

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Semiconductors in their purest form. Free from any type of impurity.

At Absolute Zero (0 *K*):

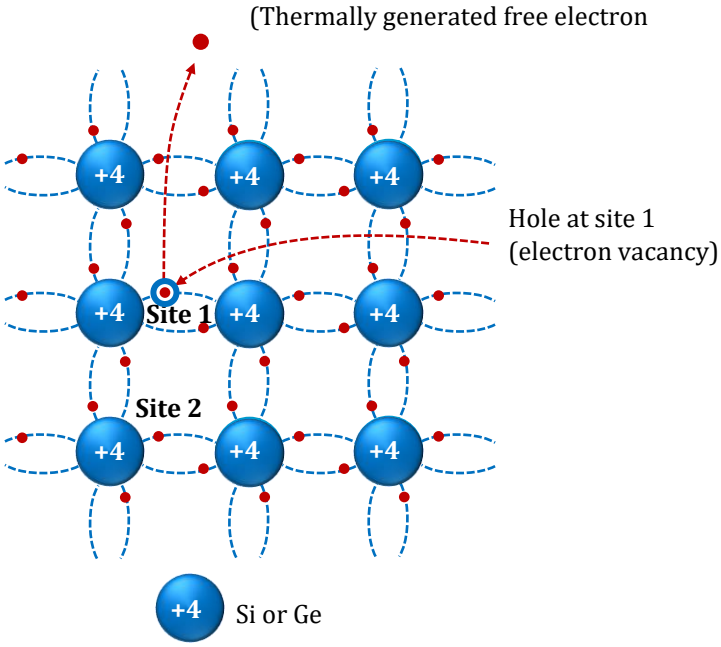


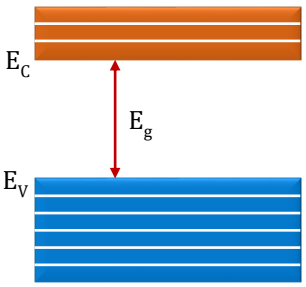
- All the covalent bonds are complete, no free electrons.
- Valence bands are completely filled.
- Conduction bands are completely empty.
- It behaves as perfect insulator.

At Room Temperature (300 K):

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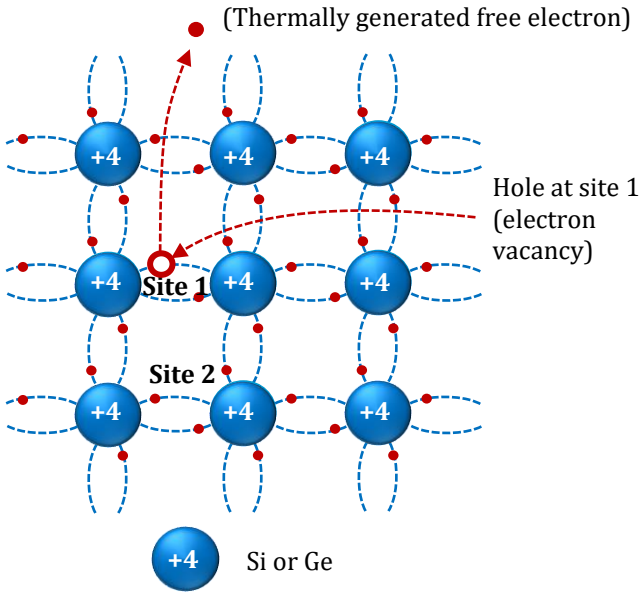
- Some of the covalent bonds are broken due to thermal energy.
- Very few free electrons.
- Valence bands are partially empty.
- Conduction bands are partially filled.





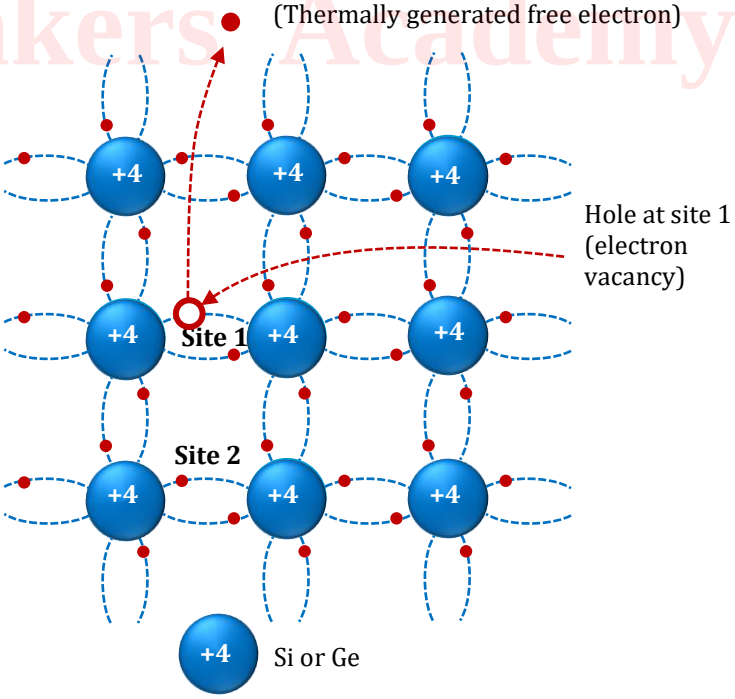
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Electron-Hole Pair (EHP):



- On receiving an additional energy, one of the electrons from a covalent band breaks and is free to move in the crystal lattice.
- While coming out of the covalent bond, it leaves behind a vacancy named 'hole'. This process is called electron-hole pair generation.

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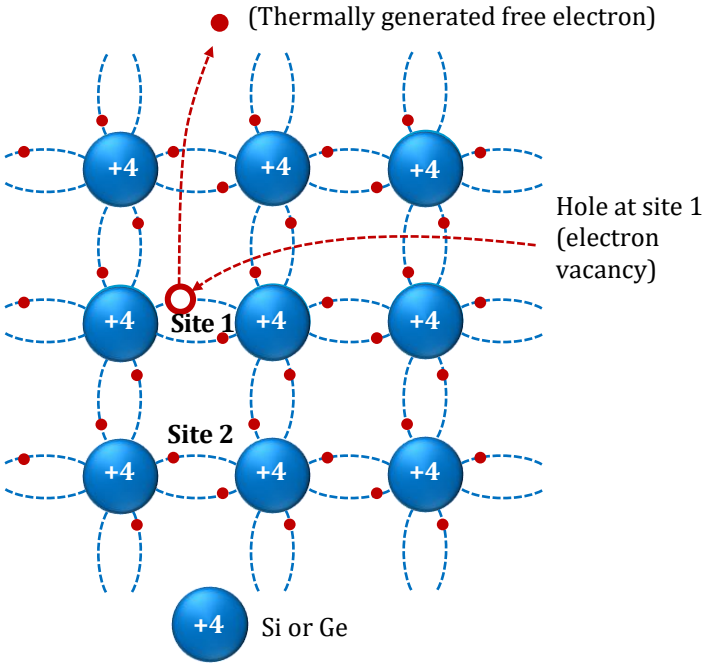


The holes move randomly in a crystal lattice. A free electron or electron from the neighboring atom can fill this vacancy and thus completing the covalent bond. This is referred to as 'electron – hole recombination'.

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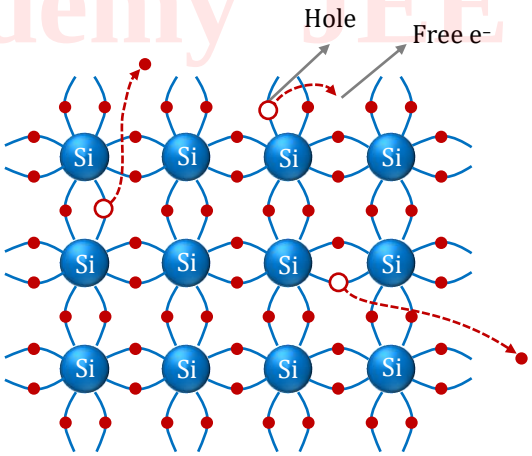
Concept of "Holes" In Semiconductors:

Due to external energy (temp. or radiation) when electron goes from valence band to conduction band (i.e. bounded electrons becomes free) a vacancy of free e⁻ creates in valence band, which has same charge as electron but positive. This positively charged vacancy is termed as hole and shown in figure.



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- These are imaginary charge carriers.
- It is deficiency of electron in VB.
- It acts as positive charge carrier.
- These are absence of e⁻ and hence there charge can be assumed to be e⁺.
- Movement of holes is due to breaking and making of covalent bonds.
- It's effective mass is more than electron.
- It's mobility is less than electron.



Note:

Hole acts as virtual charge carrier, although it has no physical significance.

In intrinsic semiconductor, the number of thermally generated electrons always equals the number of holes.

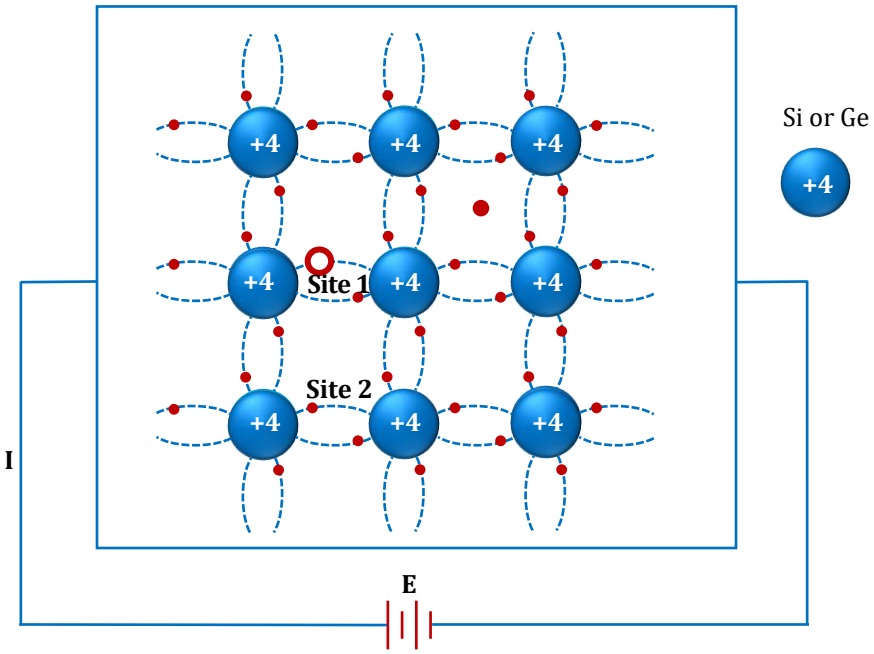
n_e = free electron concentration

n_h = hole concentration

n_e or n_h is referred to as the 'intrinsic carrier concentration' n_i.

$n_e = n_h = n_i$

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- When an intrinsic semiconductor is placed in external electric field, the charge carriers starts moving. Hence generating an electric current.

  Valence Band Current I_h

  Conduction Band Current I_e

  Total Current = $I_h + I_e$

  $I = I_e + I_h$

  $I_e = n_e e A v_e$

  $I_h = n_h e A v_h$

  So, $I = n_e e A v_e + n_h e A v_h$

  Mobility

It is ease with which charge carrier drift in electrical field. Numerical value is the drift velocity per unit electric field.

$$\mu = \frac{v_d}{E}$$

1. The electron mobility is higher than the hole mobility.

2. The resistivity / conductivity depends not only on the electron and hole densities but also on their mobilities.

3. The mobility depends relatively weakly on temperature.

$$I = \frac{V}{R} = n_e e A v_e + n_h e A v_h = eA\,(n_e v_e + n_h v_h)$$

$$\frac{VA}{\rho l} = eA\,(n_e v_e + n_h v_h)$$

$$\sigma E = e\,(n_e v_e + n_h v_h)$$

$$\sigma E = e\,(n_e \mu_e + n_h \mu_h) E$$

$$\sigma = e\,(n_e \mu_e + n_h \mu_h)$$

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Conductivity

$\sigma = e(n_e\mu_e + n_h\mu_h)$

$\sigma = \sigma_e + \sigma_h$

Where, $\sigma_e = \mu_e n_e e$

$\sigma_h = \mu_h n_h e$

- Number of electrons reaching from VB to CB at temperature T kelvin

$$n = AT^{3/2}e^{-\frac{E_g}{2kT}} = AT^{3/2}\exp\left[-\frac{E}{2kT}\right]$$

where

k = Boltzmann constant = 1.38×10^{-23} J/K T = absolute temperature

A = constant E_g = energy gap between CB and VB

- In silicon at room temperature out of 10^{12} Si atoms only one electron goes from VB to CB.
- In germanium at room temperature out of 10^{9} Ge atoms only one electron goes from VB to CB.

Effect of Temperature on Semiconductor:

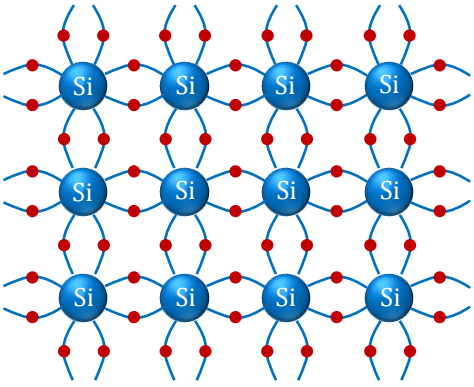
At absolute zero kelvin temperature:

At this temperature covalent bonds are very strong and there are no free electrons and semiconductor behaves as perfect insulator.

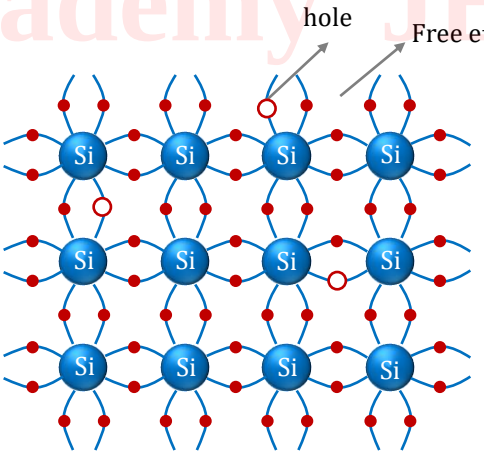
Above, absolute temperature:

With increase in temperature few valence electrons jump into conduction band and hence it behaves as poor conductor.

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At 0 K
Valence band full filled
Conduction band fully empty



At high temperature
Valence band partially empty
Conduction band partially filled

Effect of Impurity in Semiconductor

Doping is a method of addition of "desirable" impurity atoms to pure semiconductor to increase conductivity of semiconductor.

or

Doping is a process of deliberate addition of a desirable impurity atom to a pure semiconductor to modify its properties in controlled manner.

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Added impurity atoms are called dopants.

The impurity added may be ≈ 1 part per million (ppm).

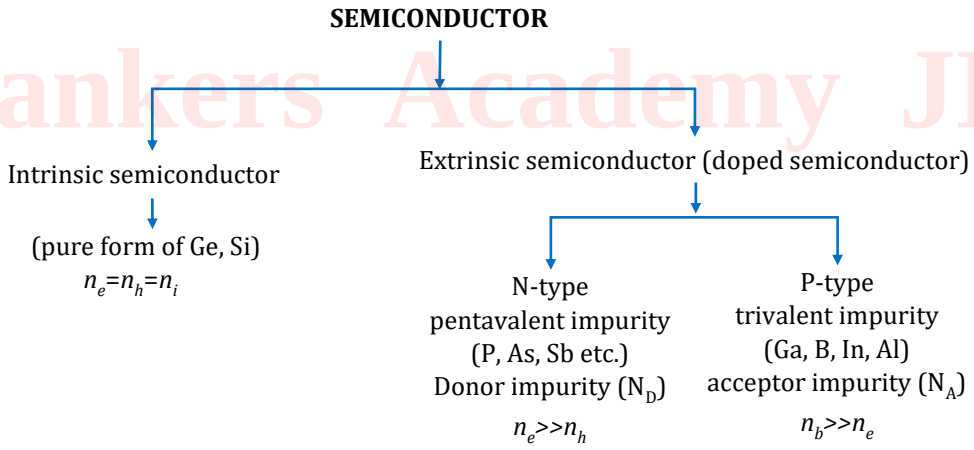
- The dopant atom should take the position of semiconductor atom in the lattice.
- The presence of the dopant atom should not distort the crystal lattice.
- The size of the dopant atom should be almost the same as that of the crystal atom.
- The concentration of dopant atoms should not be large (not more than 1% of the crystal atom).

  It is to be noted that the doping of a semiconductor increases its electrical conductivity to a great extent.
- The concentration of dopant atoms be very low, doping ratio is vary from

  Impure : pure :: $1 : 10^6$ to $1 : 10^{10}$ In general it is $1 : 10^8$
- There are two main method of doping.

  (i) Alloy method (ii) Diffusion method (The best)
- The size of dopant atom (impurity) should be almost the same as that of crystal atom. So that crystalline structure of solid remain unchanged.

Classification of Semiconductor:

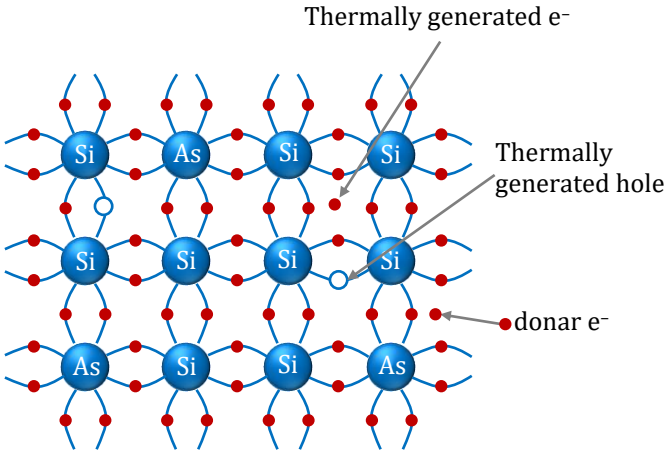


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N type semiconductor

When a pure semiconductor (Si or Ge) is doped by pentavalent impurity (*P, As, Sb, Bi*) then four electrons out of the five valence electrons of impurity take part, in covalent bonding, with four silicon atoms surrounding it and the fifth electron is set free. These impurity atoms which donate free e^- for conduction are called as Donor impurity (N_D). Due to donor impurity free e^- increases very much so it is called as "*N*" type semiconductor. By donating e^- impurity atoms get positive charge and hence known as "Immobile Donor positive Ion". In *N*-type semiconductor free e^- are called as "majority" charge carriers and "holes" are called as "minority" charge carriers.

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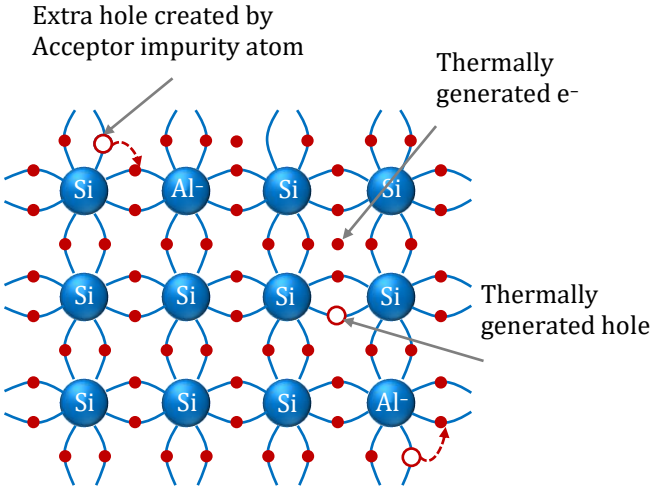


N-type semiconducting crystal

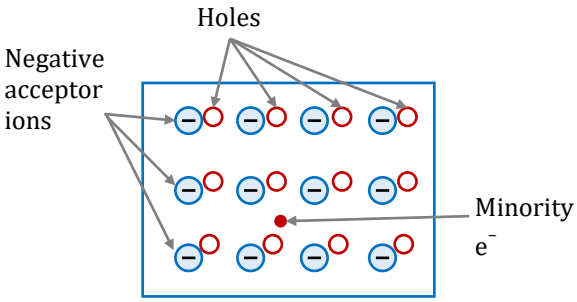
P type semiconductor

When a pure semiconductor (*Si* or *Ge*) is doped by trivalent impurity (*B, Al, In, Ga*) then outer most three electrons of the valence band of impurity take part, in covalent bonding with four silicon atoms surrounding it and except one electron from semiconductor and make hole in semiconductor. These impurity atoms which accept bonded e⁻ from valance band are called as Acceptor impurity (N_A). Here holes increase very much so it is called as "*P*" type semiconductor and impurity ions known as "Immobile Acceptor negative Ion". In *P*-type semiconductor free e⁻ are called as minority charge carries and holes are called as majority charge carriers.

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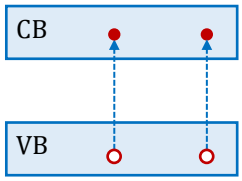
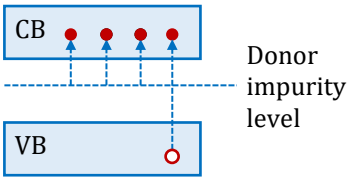
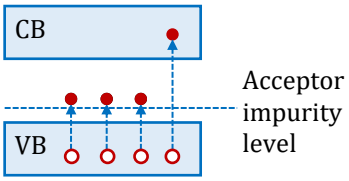
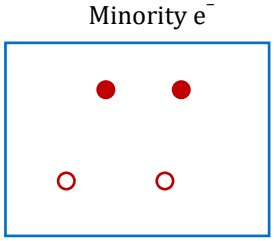
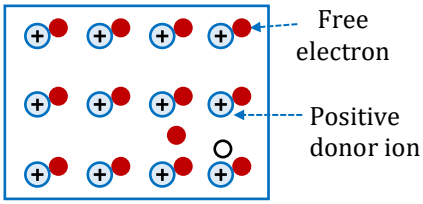
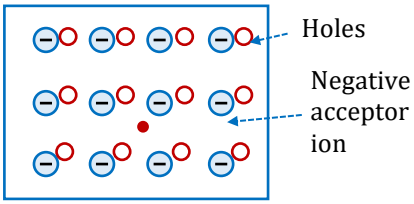


P-type semiconducting crystal



P-type semiconductor

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	Intrinsic Semiconductor	N-type (Pentavalent impurity)	P-type (Trivalent impurity)
1.			
2.			
3.	Current due to electron and hole	Mainly due to electrons	Mainly due to holes
4.	$n_e = n_h = n_i$	$n_h << n_e$ ($N_D \simeq n_e$)	$n_h >> n_e$ ($N_A \simeq n_h$)
5.	$I = I_e + I_h$	$I \simeq I_e$	$I \simeq Ih$
6.	Entirely neutral	Entirely neutral	Entirely neutral
7.	Quantity of electrons and holes are equal	Majority - Electrons Minority - Holes	Majority – Holes Minority - Electrons

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Mass action Law

In semiconductors due to thermal effect, generation of free e⁻ and hole takes place.

Apart from the process of generation, recombination also occurs simultaneously, in which free e⁻ further recombine with hole.

At equilibrium rate of generation of charge carries is equal to rate of recombination of charge carrier.

The recombination occurs due to e⁻ colliding with a hole, larger value of n_e or n_h, higher is the probability of their recombination.

Hence for a given semiconductor rate of recombination $\propto n_e \times n_h$

So, Rate of recombination = $R\, n_e \times n_h$

R = Recombination coefficient

The value of R remains constant for a solid, according to the law of thermodynamics until crystalline lattice structure remains same.

For intrinsic semiconductor $n_e = n_h = n_i$

So, Rate of recombination = $R\, n_i^2$

$R\, n_e \times n_h = R\, n_i^2$

$\Rightarrow$ $n_i^2 = n_e \times n_h$

Under thermal equilibrium, the product of the concentration $'n_e'$ of free electrons and the concentration n_h of holes is a constant and it is independent of the amount of doping by acceptor and donor impurities.

Thus, from mass action law $\boxed{n_e \times n_b = n_1^2}$

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Electron-hole Recombination

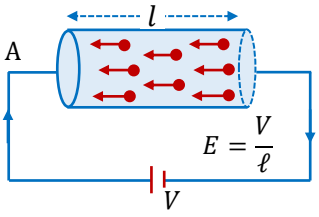
It is necessary to complete a band that electron is shared from neighboring atoms or it may also be received from conduction band. In the second case electron recombines with the hole of valance band. This process is known as electron-hole recombination.

The breaking of bands or generation of electron-hole pairs, and completion of bands due to recombination is taking place continuously.

At equilibrium, the rate of generation becomes equal to the rate of recombination, giving a fixed number of free electrons and holes.

Resistivity and Conductivity of Semiconductor

Conduction in conductor



Relation between current (I) and drift velocity (v_d)

$I = neAv_d$ n = Number of electron in unit volume

A = Cross sectional area

Current density $J = \frac{I}{A}\ amp/m^2 = ne\ v_d$ Drift velocity of electron $v_d = \mu E$

$J = ne\ \mu E$ $J = \sigma E$

Conductivity $\sigma = ne\mu = 1/\rho$ ρ = Resistivity

Mobility $\mu = \frac{v_d}{E}$

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Conduction in Semiconductor

Intrinsic semiconductor	P - type	N - type
$n_e = n_h$ $n_h >> n_e$	$n_e >> n_h$	
$J = ne\ [v_e + v_h]$	$J \cong e\ n_h\ v_h$	$J \cong e\ n_e\ v_e$
$\sigma = \frac{1}{\rho} = en[\mu_e + \mu_h]$	$\sigma = \frac{1}{\rho} \cong en_h\mu_h$	$\sigma = \frac{1}{\rho} \cong en_e\mu_e$

P-N JUNCTION

Description of P-N Junction without applied voltage or bias

Given diagram shows a P–N junction immediately after it is formed.

P region has mobile majority holes and immobile negatively charged impurity ions.

N region has mobile majority free electrons and immobile positively charged impurity ions.

Due to concentration difference diffusion of holes starts from P to N side and diffusion of e^- s starts N to P side.

Due to this a layer of only positive (in N side) and negative (in P–side) started to form which generate an electric field (N to P side) which oppose diffusion process, during diffusion magnitude of electric field increases due to this diffusion it gradually decreased and ultimately stope.

The layer of immobile positive and negative ions, which have no free electrons and holes called as **depletion layer** as shown in diagram.

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- **Width of depletion layer $\cong 10^{-6}\ m$**
  (a) As doping increases depletion layer decreases
  (b) As temperature is increased depletion layer also increases.
  (c) *P-N* junction $\rightarrow$ nonohmic, due to nonlinear relation between I and V.
- **Potential Barrier or contact potential**
  $Ge \longrightarrow 0.3\ V$ $\qquad$ $Si \longrightarrow 0.7\ V$
- Electric field, produce due to potential barrier $E = \frac{V}{d} = \frac{0.5}{10^{-6}} \Rightarrow E \cong 10^5\ V/m$

  This field prevents the respective majority carrier from crossing barrier region

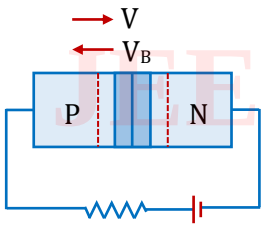
Diffusion and Drift Current

(1) Diffusion current – *P* to *N* side
(2) Drift current – *N* to *P* side
If there is no biasing diffusion current = drift current
So total current is zero.

Behaviour of P–N Junction with An External Voltage Applied or Bias

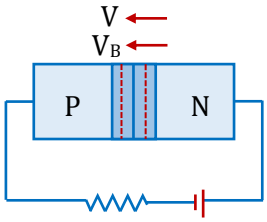
Forward Bias

If we apply a voltage *"V"* such that *P*–side is positive and *N*–side is negative as shown in diagram.
The applied voltage is opposite to the junction barrier potential. Due to this effective potential barrier decreases, junction width also decreases, so more majority carriers will be allowed to flow across junction. It means the current flow in principally due to majority charge carriers and it is in the order of *mA* called as forward Bias.



Reverse Bias

If we apply a voltage "*V*" such that *P*–side is negative and *N*–side is positive as shown in diagram. The applied voltage is in same direction as the junction barrier potential. Due to this effective potential barrier increase junction, width also increases, so no majority carriers will be allowed to flow across junction.



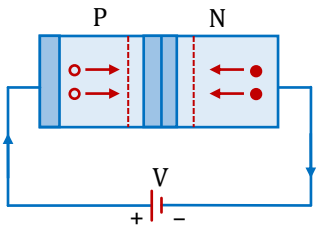
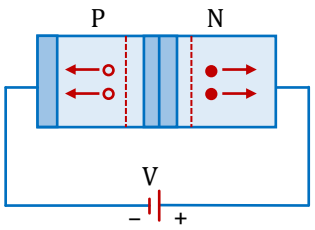
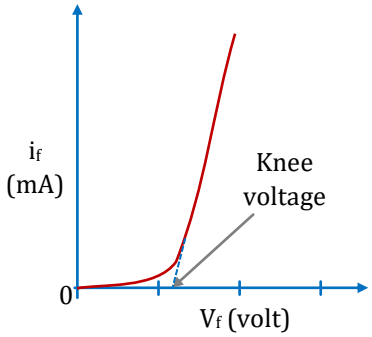
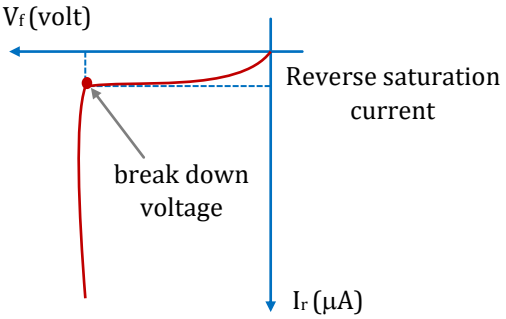
Only minority carriers will be drifted. It means the current flow in principally due to minority charge carriers and is very small (in the order of μA). This bias is called as reversed Bias.

- In reverse bias, the current is very small and nearly constant with bias (termed as reverse saturation current). However interesting behavior results in some special cases if the reverse bias is increased further beyond a certain limit, above particular high voltage breakdown of depletion layer started.

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- Breakdown of a diode is of following two types:
  (i) Zener breakdown
  (ii) Avalanche breakdown

Comparison between Forward Bias and Reverse Bias

Forward Bias	Reverse Bias
P → positive N → negative	P → negative N → positive
	
1. Potential Barrier reduces	**1.** Potential Barrier increases.
2. Width of depletion layer decreases	**2.** Width of depletion layer increases.
3. *P-N* jn. provide very small resistance	**3.** *P-N* jn. provide high resistance
4 Forward current flows in the circuit	**4.** Very small current flows.
5. Order of forward current is milli ampere.	**5.** Order of current is micro ampere for Ge or Nano ampere for Si.
6. Current flows mainly due to majority carriers.	**6.** Current flows mainly due to minority carriers.
7. Forward characteristic curves.	**7.** Reverse characteristic curve
	
8. Forward resistance $R_f = \frac{\Delta V_f}{\Delta I_f} \cong 100\Omega$	**8.** Reverse resistance $R_r = \frac{\Delta V_r}{\Delta I_r} \cong 10^6\Omega$
9. Order of knee or cut in voltage *Ge* → 0.3 *V* *Si* → 0.7 *V* Special point: Generally, $\frac{R_r}{R_f} = 10^3$: 1 for Ge	**9.** Breakdown voltage *Ge* → 25 *V* *Si* → 35 *V* $\frac{R_r}{R_f} = 10^4$: 1 for Si

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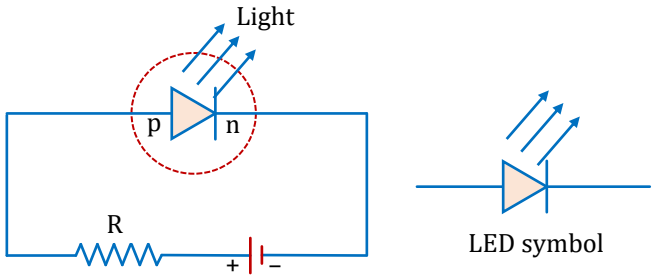
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Light Emitting Diode (LED):

A light emitting diode is simply a forward biased *p-n* junction which emits spontaneous light radiation. When forward bias is applied, the electron and holes at the junction recombine and energy released is emitted in the form of light. for visible radiation phosphorus doped GaAs is commonly used. The advantages of LEDs are:

(i) Low operational voltage and less power.
(ii) Fast action with no warm up time.
(iii) Emitted light is nearly monochromatic.
(iv) They have long life.

I-V characteristics of LED are similar to that of Si junction diode but the threshold voltages are much higher and slightly different for each color. The reverse breakdown voltages of LED's are very low, about 5 *V*.



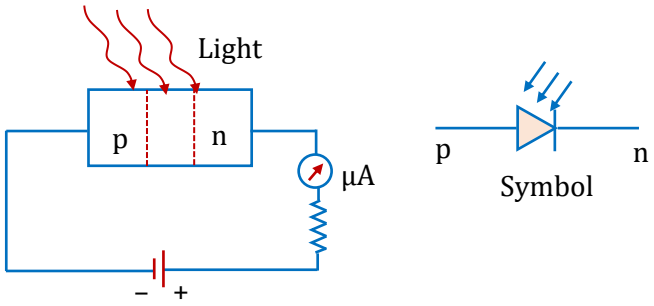
Photodiode:

It is a reversed-biased p-n junction, illuminated by radiation. When p-n junction is reversed biased with no current, a very small reverse saturated current flows across the junction called the dark current. When the junction is illuminated with light, electron-hole pairs are created at the junction, due to which additional current begins to flow across the junction; the current is solely due to minority charge carriers.

(1) A photodiode is used in reverse bias, although in forward bias current is more than current in reverse bias because in reverse bias it is easier to observe change in current with change in light intensity.

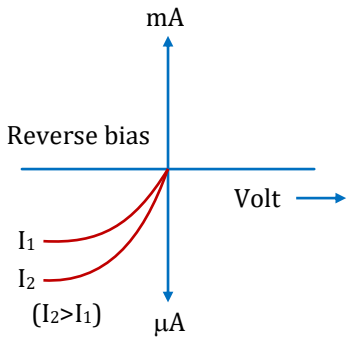
(2) Photodiode is used to measure light intensity because reverse current increases with increase of intensity of light.

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(a) **(b)**

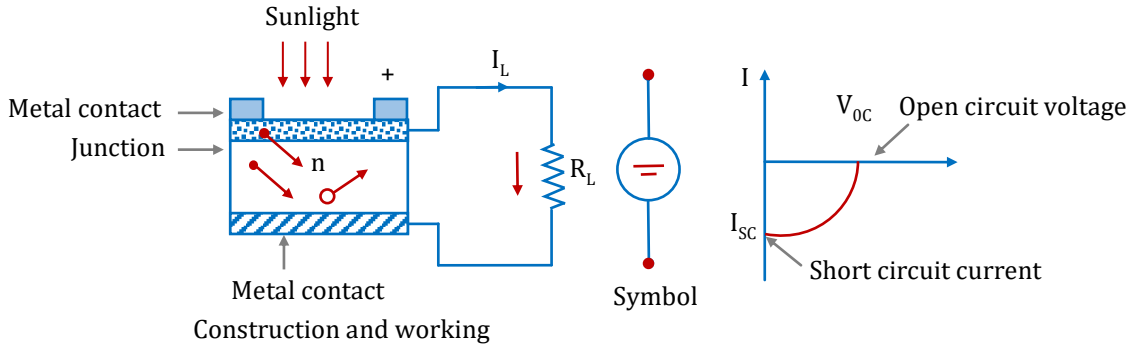
The characteristic curves of a photodiode for two different illumination I_1 and I_2 ($I_2 > I_1$) are shown in figure.



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Solar Cell

A solar cell is a junction diode which converts light energy 'into electrical energy. A p-n junction solar cell consists of a large junction with no external biasing.



(a) **(b)**

The surface layer of p-region is made very thin so that the incident photons may easily penetrate to reach the junction which is the active region. In an operation in the photovoltaic mode (i.e., generation of voltage due to bombardment of optical photons); the materials suitable for photocells are silicon (Si), gallium arsenide (GaAs), cadmium sulphide (CdS) and cadmium selenide (CdSe).

Working:

When photons of energy greater than band gap energy ($hv > E_g$) are made incident on the junction, electron-hole pairs are created which move in opposite directions due to junction field. These are collected at two sides of junction, thus producing photo-voltage; this gives rise to photocurrent. The characteristic curve of solar cell is shown in fig. Solar cells are used in satellites to recharge their batteries.

Pankaj's Academy JEE

Reverse breakdown:

If the reverse bias voltage is made too high, the current through the PN junction increases rapidly at Y_z. The voltage at which this happens is called **breakdown voltage** or **Zener voltage.**

There two mechanism which causes this breakdown. One is called avalanche breakdown and other is called **Zener breakdown.**

Zener breakdown:

When reverse bias is increased the electric field at the junction also increases. At some stage the electric field becomes so high that it breaks the covalent bonds creating electron, hole pairs, thus a large number of carriers are generated. This causes a large current to flow. This mechanism is known as Zener breakdown.

Avalanche breakdown:

At high reverse voltage, due to high electric field, the minority charge carriers, while crossing the junction acquires very high velocities. These by collision breaks down the covalent bonds, generating more carriers. A chain reaction is established, giving rise to high current. This mechanism is called **avalanche breakdown.**

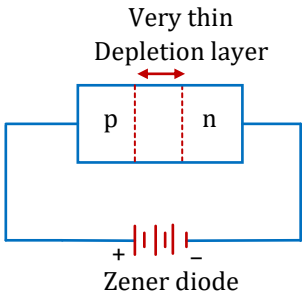
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Zener Diode:

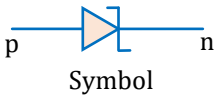
A Zener diode is a specially designed heavily doped p-n junction, having a very thin depletion layer and having a very sharp breakdown voltage. It is always operated in breakdown region. Its breakdown voltage V_z is less than $6V$.

Zener Diode as a Voltage Regulator:

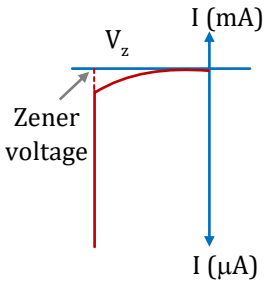
Zener diode may be used as a voltage regulator. The circuit of Zener-diode is shown in figure.



(a)



(b)

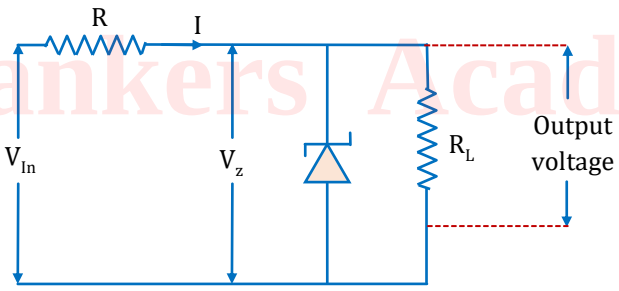


(c) Characteristics

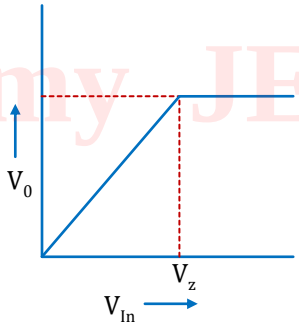
In breakdown region the equation:

$$V_0 = V_z = V_{in} - RI$$

Clearly, when the input voltage exceeds Zener voltage to keep the voltage regularity, the extra input voltage appears across series resistance R. The voltage regulation curve is shown in figure.



(d)



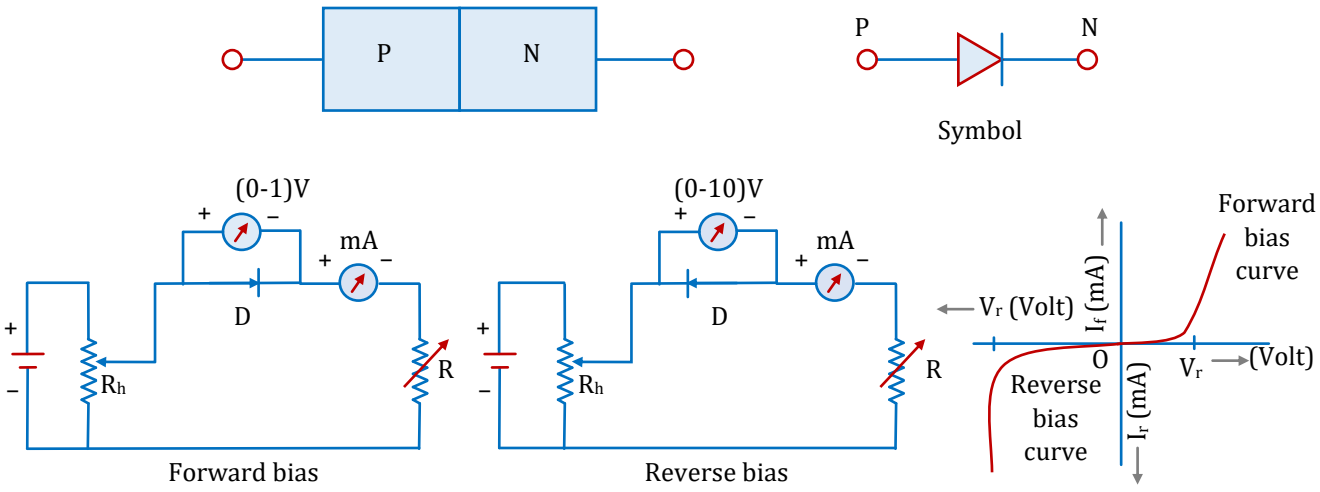
(e)

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Zener Break down	**Avalanche Break down**
Where covalent bonds of depletion layer, its self-break, due to high electric field of very high Reverse bias voltage.	Here covalent bonds of depletion layers are broken by collision of "Minorities" which acquire high kinetic energy from high electric field of very-very high reverse bias voltage.
This phenomenon predominant	This phenomenon predominant
(i) At lower voltage after "break down"	(i) At high voltage after breakdown
(ii) In P – N having "High doping"	(ii) In P – N having "Low doping"
(iii) P – N Jn. having thin depletion layer	(iii) P – N Jn. having thick depletion layer
Here P – N not damage permanently	Here P – N damage permanently due to
"In D.C voltage stabilizer Zener phenomenon is used".	"Heating effect" due to abruptly increment of minorities during repetitive collisions.

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Characteristic Curve Of P-N Junction Diode



In forward bias when voltage is increased from 0*V* is steps and corresponding value of current is measured, the curve comes as *OB* of figure. We may note that current increase very sharply after a certain voltage knee voltage. At this voltage, barrier potential is completely eliminated and diode offers a low resistance. In reverse bias a microammeter has been used as current is very small. When reverse voltage is increased

from 0*V* and corresponding values of current measured the plot comes as *OCD*. We may note that reverse current is almost constant hence called reverse saturation current. It implies that diode resistance is very high. As reverse voltage reaches value V_B, called breakdown voltage, current increases very sharply.

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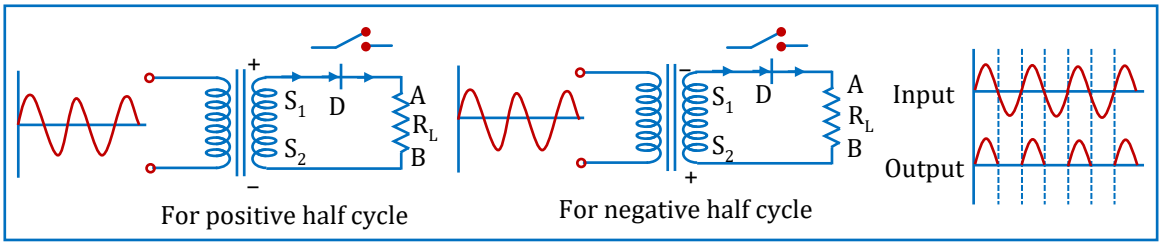
For Ideal Diode

In forward bias		In reverse bias	
'ON' switch	$R_f = 0$	'OFF' switch	$R_f = \infty$

Rectifier

It is device which is used for converting alternating current into direct current.

Half Wave Rectifier



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During the first half (positive) of the input signal, let S_1 is at positive and S_2 is at negative potential. So, the PN junction diode D is forward biased. The current flows through the load resistance R_L and output voltage is obtained.

During the second half (negative) of the input signal, S_1 and S_2 would be negative and positive respectively. The PN junction diode will be reversed biased. In this case, practically no current would flow through the load resistance. So, there will be no output voltage.

Thus, corresponding to an alternating input signal, we get a unidirectional pulsating output as shown.

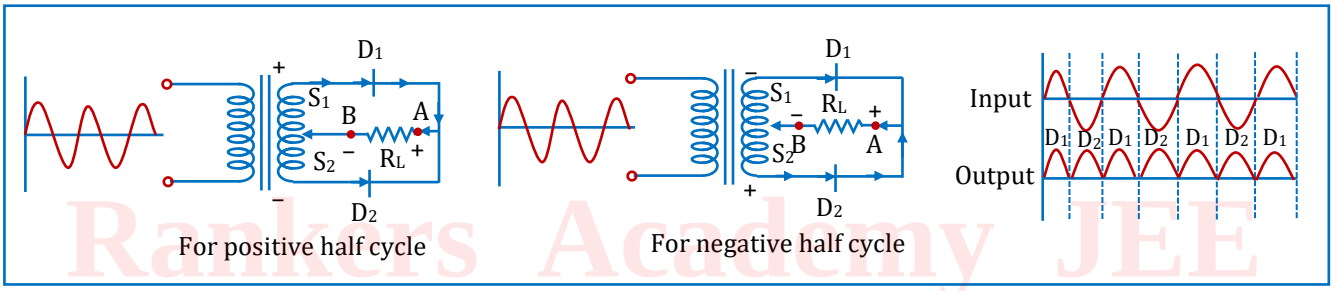
Peak Inverse Voltage (PIV)

In half wave rectifier PIV = maximum voltage across secondary coil of transformer (V_s)

= Peak value of output (V_m)

Full Wave Rectifier

When the diode rectifies the whole of the AC wave, it is called full wave rectifier. Figure shows the experimental arrangement for using diode as full wave rectifier. The alternating signal is fed to the primary a transformer. The output signal appears across the load resistance R_L.



During the positive half of the input signal:

Let S_1 positive and S_2 negative.

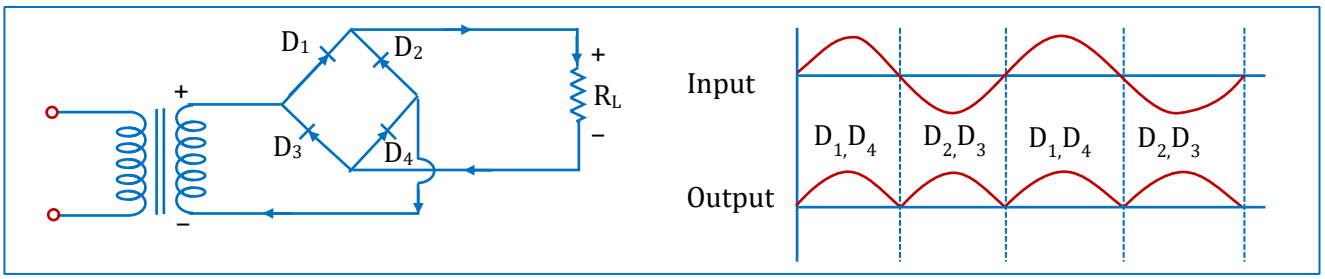
In this case diode D_1 is forward biased and D_2 is reverse biased. So only D_1 conducts and hence the flow of current in the load resistance R_L is from A to B.

During the negative half of the input signal:

Now S_1 is negative and S_2 is positive. So D_1 is reverse-biased and D_2 is forward biased. So only D_2 conducts and hence the current flows through the load resistance R_L from A to B.

It is clear that whether the input signal is positive or negative, the current always flows through the load resistance in the same direction and full wave rectification is obtained.

Bridge Rectifier



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During positive half cycle

D_1 and D_4 are forward biased $\rightarrow$ on switch

D_2 and D_3 are reverse biased $\rightarrow$ off switch

In bridge rectifier peak inverse voltage $PIV = V_s = V_m$

During negative half cycle

D_2 and D_3 are forward biased $\rightarrow$ on switch

D_1 and D_4 are reverse biased $\rightarrow$ off switch

Form Factor

$$F = \frac{I_{rms}}{I_{dc}} \text{ or } \frac{E_{rms}}{E_{dc}}$$

For full wave rectifier $F = \frac{\pi}{2\sqrt{2}}$

For half wave rectifier $F = \frac{\pi}{2}$

Ripple and Ripple Factor

In the output of rectifier some A.C. components are present. They are called ripple & their measurement is given by a factor known as ripple factor. For a good rectifier ripple factor must be very low.

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Total output current

$I_{rms} = \sqrt{I_{ac}^2 + I_{dc}^2}$ Where I_{ac} = *rms* value of AC component present in output

Ripple factor, $r = \frac{I_{ac}}{I_{dc}} \Rightarrow r = \sqrt{\frac{I_{rms}^2}{I_{dc}^2} - 1} = \sqrt{F^2 - 1}$

Rectifier efficiency $\eta = \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (R_F + R_L)}$

Half wave rectifier

$$\eta = \frac{0.406}{1 + \frac{R_f}{R_L}}$$

If $\frac{R_f}{R_L} << 1$, then $\eta = 40.6\%$

Special Note If $R_f = R_L$

$\eta = 20.3\%$

Full wave rectifier or bridge wave rectifier

$$\eta = \frac{0.812}{1 + \frac{R_f}{R_L}}$$

If $\frac{R_f}{R_L} << 1$, then $\eta = 81.2\%$

Special Note If $R_f = R_L$

$\eta = 40.6\%$

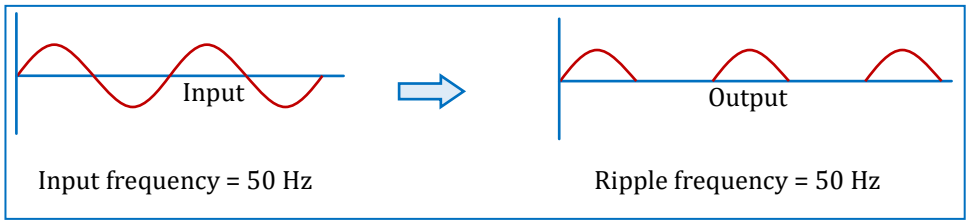
Note:

In bridge full wave rectifier R_f is two times of resistance of P-N jn. diode in *FB*.

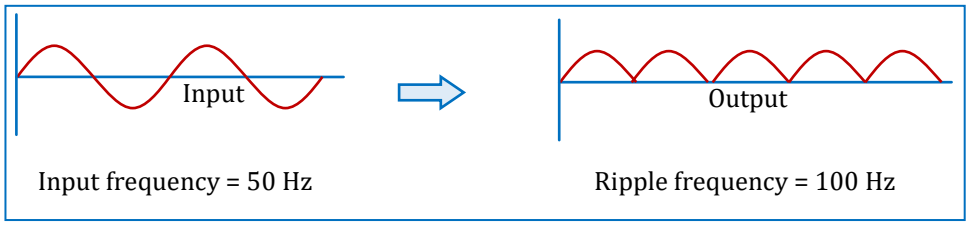
For More Join: @IITJEE_Advanced

Ripple Frequency

(i) For half wave rectifier

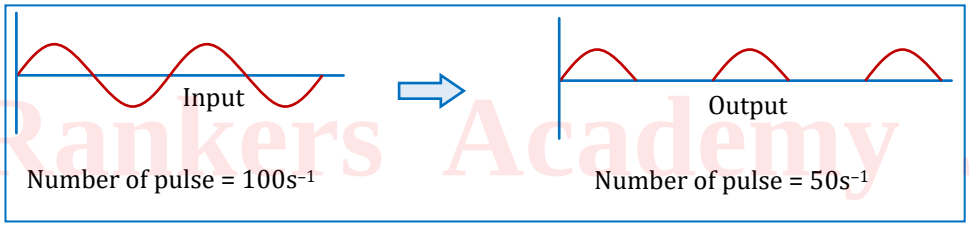


(ii) For full wave rectifier



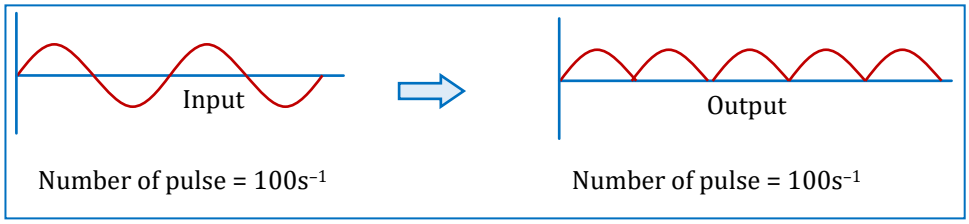
Pulse

(i) For half wave rectifier



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(ii) For full wave rectifier



Comparison Between Average Rectifiers			
	Half-wave	**Centre-tap**	**Full-wave Bridge**
Number of Diodes	1	2	4
Transformer necessary	No	Yes	No
Peak secondary voltage	V_s	V_s	V_s
Peak Inverse Voltage (when peak of output = V_m)	$V_s = V_m$	$V_s = 2V_m$	$V_s = V_m$

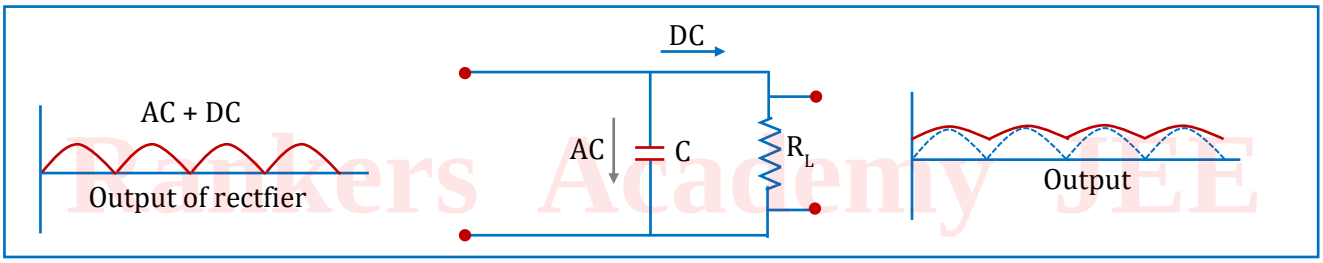
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Comparison Between Average Rectifiers			
	Half-wave	**Centre-tap**	**Full-wave Bridge**
Peak load Current, I_m	$\frac{V_{in}}{r_d + R_L}$	$\frac{V_{in}}{r_d + R_L}$	$\frac{V_{in}}{2r_d + R_L}$
RMS Current, I_{rms}	$\frac{I_m}{2}$	$\frac{I_m}{\sqrt{2}}$	$\frac{I_m}{\sqrt{2}}$
DC current, I_{dc}	$\frac{I_m}{\pi}$	$\frac{2I_m}{\pi}$	$\frac{2I_m}{\pi}$
Ripple factor, r	1.21	0.482	0.482
Rectification efficiency (max)	40.6%	81.2%	81.2%
Ripple frequency (when input = 50 Hz)	50 Hz	100 Hz	100 Hz

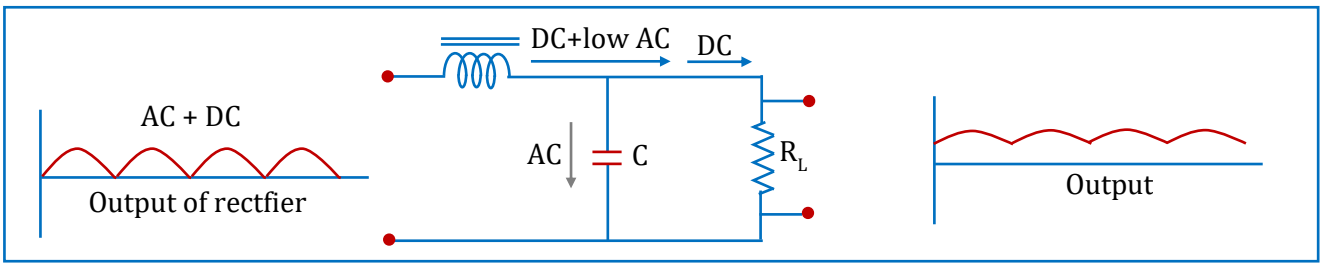
Filter Circuit

To reduce A.C. Components

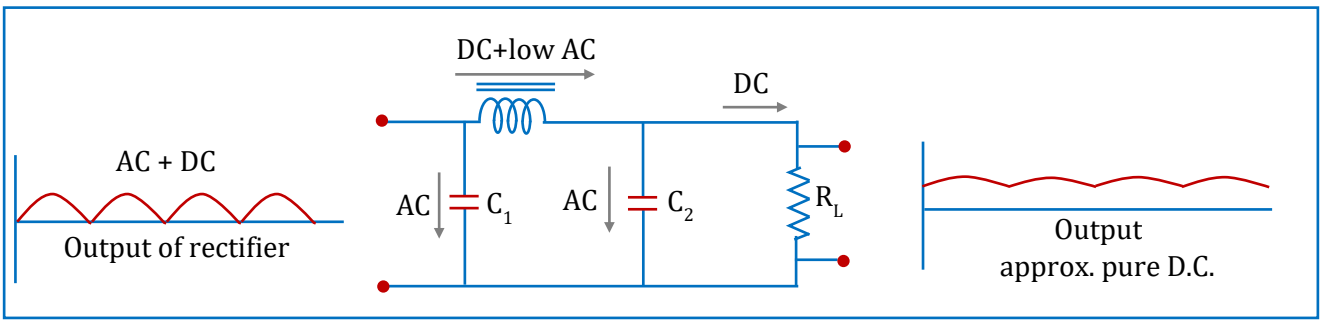
Capacitor Filter



L – C Filter



π - Filter (Best Filter)



For More Join: @IITJEE_Advanced

Diode laser

It is interesting form of LED in which special construction helps to produce stimulated radiation as in laser.

LOGIC GATES

Introduction:

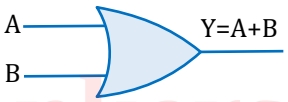
- A logic gate is a digital circuit which is based on certain logical relationship between the input and the output voltages of the circuit.
- The logic gates are built using the semiconductor diodes and transistors.
- Each logic gate is represented by its characteristic symbol.
- The operation of a logic gate is indicated in a table, known as truth table. This table contains all possible combinations of inputs and the corresponding outputs.
- A logic gate is also represented by a Boolean algebraic expression. Boolean algebra is a method of writing logical equations showing how an output depends upon the combination of inputs. Boolean algebra was invented by George Boole.

Basic Logic Gates

There are three basic logic gates. They are (1) OR gate (2) AND gate, and (3) NOT gate.

- **The OR gate:** The output of an OR gate attains the state 1 if one or more inputs attain the state 1.

  Logic symbol of OR gate



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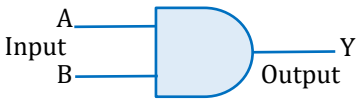
  The Boolean expression of OR gate is $Y = A + B$, read as Y equals A 'OR' B.

  Truth table of a two-input OR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

- **The AND gate :-** The output of an AND gate attains the state 1 if and only if all the inputs are in state 1.

  Logic symbol of AND gate



  The Boolean expression of AND gate is $Y = A.B$

  It is read as Y equals A 'AND' B

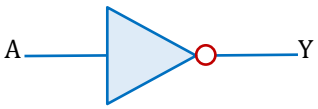
  Truth table of a two-input AND gate

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

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- **The NOT gate:** The output of a NOT gate attains the state 1 if and only if the input does not attain the state 1.

  Logic symbol of NOT gate



  The Boolean expression is $Y = \bar{A}$, read as Y equals NOT A.

  Truth table of NOT gate

A	Y
0	1
1	0

Combination of gates:

The three basis gates (OR, AND and NOT) when connected in various combinations give us logic gates such as NAND, NOR gates, which are the universal building blocks of digital circuits.

- **The NAND gate:**

  Logic symbol of NAND gate



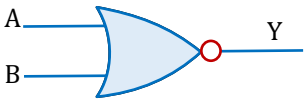
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  The Boolean expression of NAND gate is $Y = \overline{A.B}$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

- **The NOR gate:**

  Logic symbol of NOR gate



  The Boolean expression of NOR gate is $Y = \overline{A + B}$

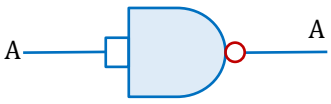
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

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Universal gates:

The NAND or NOR gate is the universal building block of all digital circuits. Repeated use of NAND gates (or NOR gates) gives other gates. Therefore, any digital system can be achieved entirely from NAND or NOR gates. We shall show how the repeated use of NAND (and NOR) gates will gives us different gates.

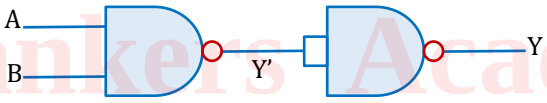
- **The NOT gate from a NAND gate: -** When all the inputs of a NAND gate are connected together, as shown in the figure, we obtain a NOT gate



Truth table of a single input NAND gate

A	B=(A)	Y
0	0	1
1	1	0

- **The AND gate from a NAND gates:-** If a NAND gate is followed by a NOT gate (i.e., a single input NAND gate), the resulting circuit is an AND gate as shown in figure and truth table given show how an AND gate has been obtained from NAND gates.

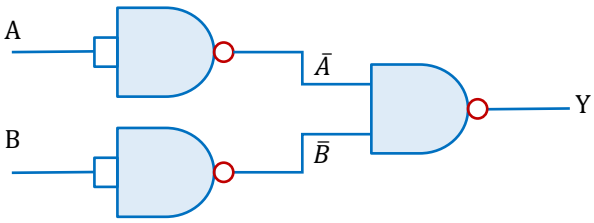


Truth table

A	B	Y'	Y
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

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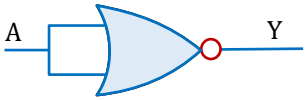
- **The OR gate from NAND gates: -** If we invert the inputs *A* and *B* and then apply them to the NAND gate, the resulting circuit is an OR gate.



Truth table

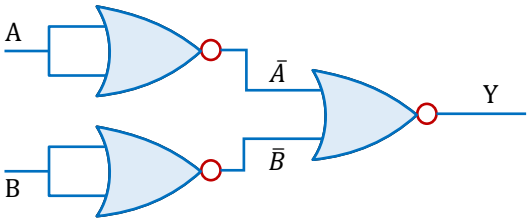
A	B	$\bar{A}$	$\bar{B}$	Y
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

- **The NOT gate from NOR gates:-** When all the inputs of a NOR gate are connected together as shown in the figure, we obtain a NOT gate

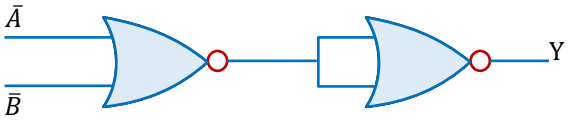


- **The AND gate from NOR gates:-** If we invert the inputs *A* and *B* and then apply them to the NOR gate, the resulting circuit is an AND gate.

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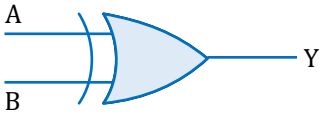
- **The OR gate from NOR gate:-** If a NOR gate is followed by a single input NOR gate (NOT gate), the resulting circuit is an OR gate.



XOR and XNOR gates:

- **The Exclusive - OR gate (XOR gate): -** The output of a two-input XOR gate attains the state 1 if one and only one input attains the state 1.

  Logic symbol of XOR gate



The Boolean expression of XOR gate is or $Y = A \oplus B$

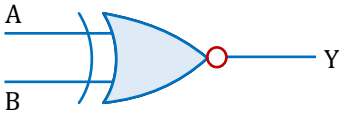
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Truth table of a XOR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

- **Exclusive - NOR gate (XNOR gate): -** The output is in state 1 when it's both inputs are the same that is, both 0 or both 1.

  Logic symbol of XNOR gate



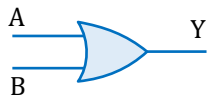
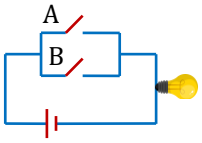
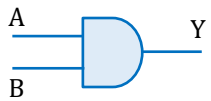
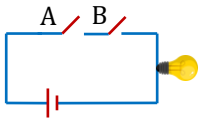
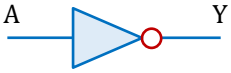
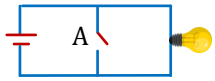
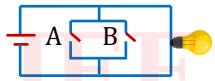
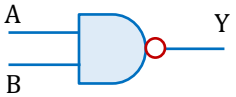
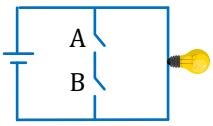
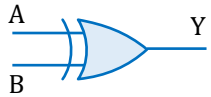
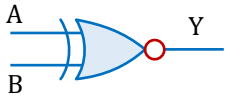
The Boolean expression of XNOR gate is or $A \odot B$

Truth table of a XNOR gate

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

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Summary of Logic Gates

Names	Symbol	Boolean Expression	Truth Table	Electrical Analogue
OR		$Y = A + B$	A B Y 0 0 0 0 1 1 1 0 1 1 1 1	
AND		$Y = A.B$	A B Y 0 0 0 0 1 0 1 0 0 1 1 1	
NOT or Inverter		$Y = \bar{A}$	A Y 0 1 1 0	
NOR (OR+NOT)		$Y=\overline{A+B}$	A B Y 0 0 1 0 1 0 1 0 0 1 1 0	
NAND (AND+ NOT)		$Y=\overline{A.B}$	A B Y 0 0 1 0 1 1 1 0 1 1 1 0	
XOR (Exclusive OR)		$Y=A \oplus B$ or $Y=\bar{A}.B + A.\bar{B}$	A B Y 0 0 0 0 1 1 1 0 1 1 1 0	
XNOR (Exclusive NOR)		$Y=A \odot B$ or $Y=A.B + \bar{A}.\bar{B}$ or $Y=\overline{A \oplus B}$	A B Y 0 0 1 0 1 0 1 0 0 1 1 1	

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CHARACTERISTIC CURVES OF A P-N JUNCTION

EXPERIMENT # 1

Aim

To study the static and dynamic curves of a p-n junction diode in forward bias and to determine its static and dynamic resistances.

Requirements

A p-n junction diode, a three-volt battery, a high resistance rheostat, 0-3-volt voltmeter, one milliammeter, one-way key and connecting wires.

Theory

When a junction diode is forward biased, a forward current is produced which increases with increase in bias voltage. This increase is not proportional.

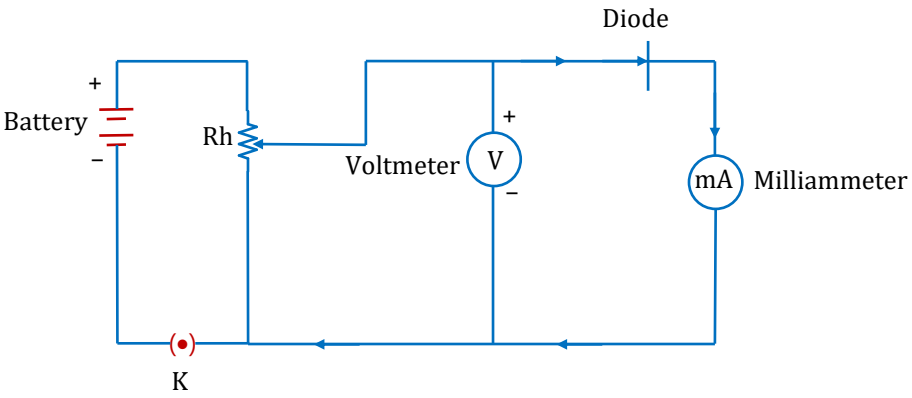
The ratio of forward bias voltage (V) and forward current (I) is called the static resistance of semiconductor diode, i.e., $R = \frac{V_F}{I_F}$.

In case of a varying bias voltage and varying forward current, the ratio of change in forward bias voltage (ΔV) and corresponding change in forward current (ΔI) is called the dynamic resistance $\left(r = \frac{\Delta V_F}{\Delta I_F}\right)$.

To find the static and dynamic resistance of semiconductor diode, a graph has to be plotted between forward bias voltage (V) and forward bias current (I). This graph is called the characteristic curve of semiconductor diode.

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Procedure



(a) Make the connections as shown in above figure.

(b) Keep the moving contact of the rheostat to the minimum and insert the key K. Voltmeter and milliammeter will show a zero reading.

(c) Move the contact towards the positive to apply the forward bias voltage V = 0.1 V. The current remains zero.

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(d) Increase the forward bias voltage to 0.3 *V* in steps. The current will still be zero. (This is due to the junction potential barrier of 0.3 *V*).

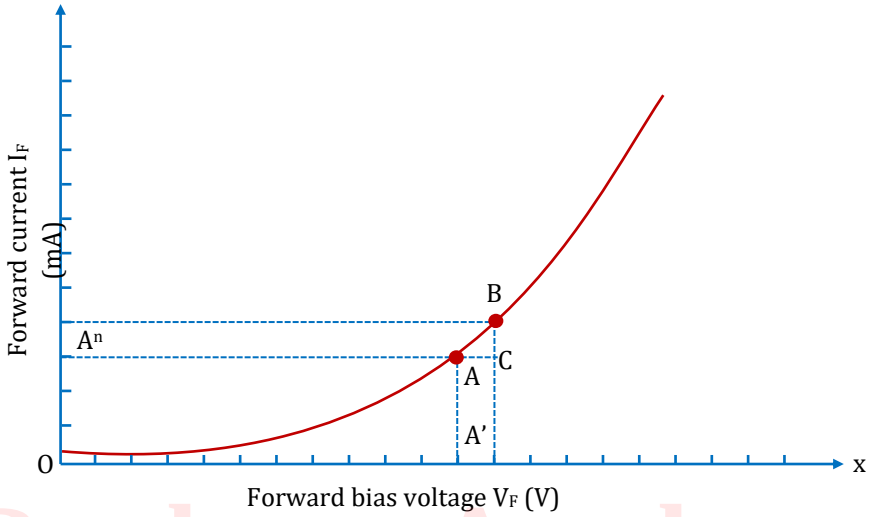
(e) Increase *V* to 0.4 *V*. Record the current.

(f) Increase *V* in step of 0.2 and note the corresponding current.

(g) At *V* = 2.4 *V*. The current increases suddenly. This represents the forward breakdown stage.

(h) Draw a graph of *I* on *y*-axis and *V* on *x*-axis. The graph will be as shown in figure.

Graph



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Calculations

(i) For static resistance (*R*) $R = \frac{V_F}{I_F}$

From the graph $R = \frac{OA'}{OA}$ = ..., *ohm*

Diode used.....(specify the code)

(ii) For dynamic resistance (*r*) $r = \frac{\Delta V_F}{\Delta I_F}$

$\therefore$ From the graph $r = \frac{AC}{BC}$ *ohms*

Precautions

(i) Make all connections neat, clean and tight.

(ii) Key should be used in circuit and opened when the circuit is not in use.

(iii) Avoid applying forward bias voltage beyond breakdown.

Possible Sources of Errors

(i) The connection may not be tight.

(ii) The junction diode may be faulty.

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CHARACTERISTIC CURVES OF A ZENER DIODE

EXPERIMENT # 2

Aim

To draw the characteristic curves of a Zener diode and to determine its reverse breakdown voltage.

Apparatus

A Zener diode (with reverse breakdown voltage of 6 *V*), a ten-volt battery, a rheostat, two voltmeters (range 0, 10 *V*), one milliammeter, one 20Ω resistance, one-way key, connecting wires.

Theory

Zener diode is a semiconductor diode in which the *n*-type sections are heavily doped, this heavy doping results in a low value of reverse breakdown voltage.

The reverse breakdown voltage of Zener diode is called Zener voltage (V_Z). The reverse current that results after the breakdown, is called Zener current (I_z).

The circuit diagram is given in figure

V_i = Input voltage

V_0 = Output voltage

R_i = Input resistance

I_i = Input current

I_Z = Zener diode current

I_L = Load current

$I_L = I_i - I_Z$

$V_0 = V_i - R_i I_i$

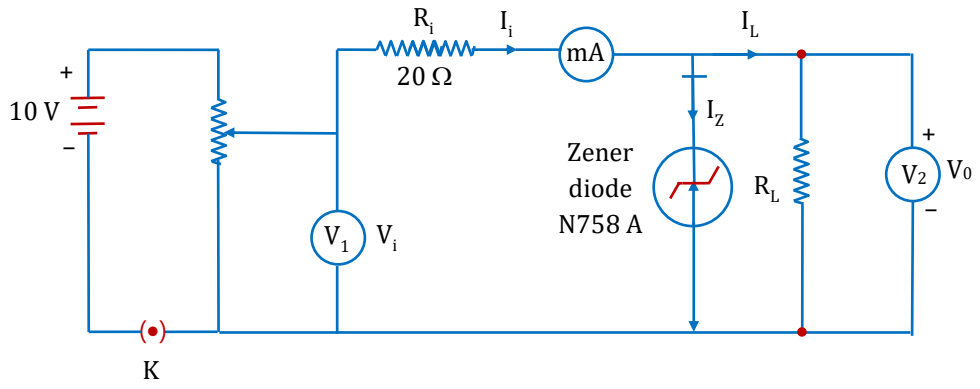
$V_0 = R_L I_L$

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Initially as V_i increases, I_i increases hence V_0 increases linearly. At break-down, increase of V_i increases I_i by large amount, so that $V_0 = V_i - R_i I_i$ becomes constant.

This constant value of V_0 Which is the reverse breakdown voltage, is called Zener voltage.

Circuit Diagram



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Procedure

(a) Make the connections as shown in figure above making sense that Zener diode is reverse biased.

(b) Bring the moving contact to rheostat to the minimum and insert the key K. Voltmeter and ammeter will read zero.

(c) Move the contact a little towards positive end to apply some reverse bias voltage (V_i). Milliammeter reading remains zero.

(d) As V_i is further increased, I_i starts increasing and V_0 becomes less than V_i. Note the values of V_i, V_0 and I_i.

(e) Keep increasing V_i in small steps of 0.5 V. Note the corresponding values of I_i and V_0.

(f) At one stage as V_i is increased, I_i increases by large amount and V_0 does not increase. This is reverse break down situation.

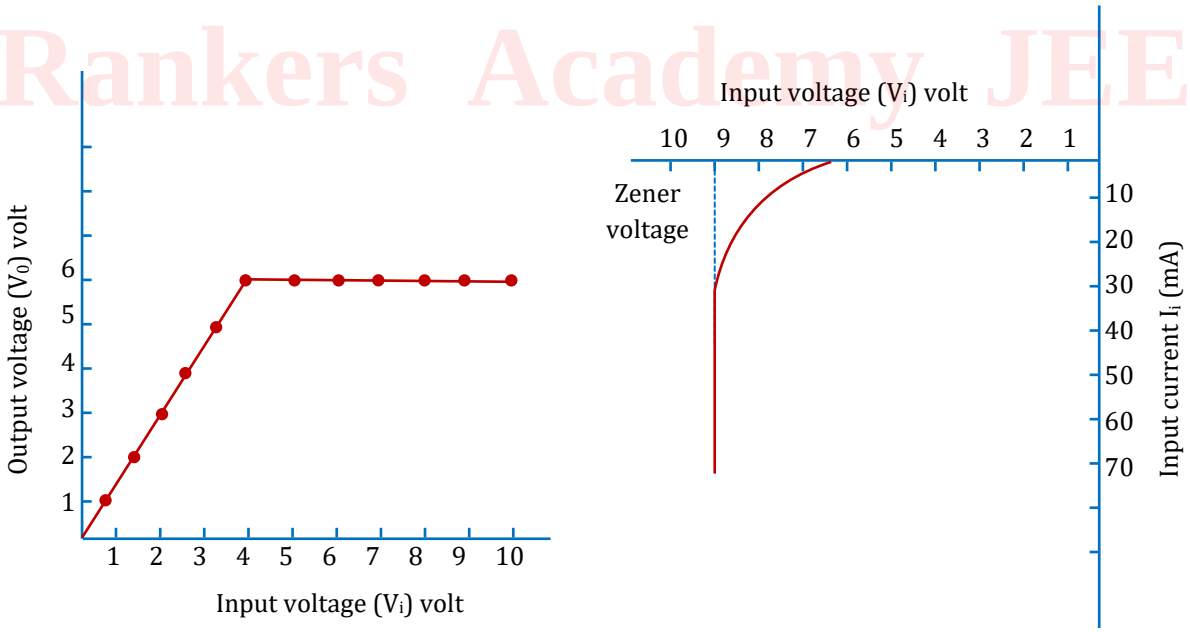
(g) As V_i is increased further, I_i will increase keeping V_0 constant. Record your observation in tabular column.

(h) Draw graph of output voltage V_0 along y-axis and input voltage along x-axis. The graph will be as shown in figure.

(i) Draw graph of input current along y-axis and input voltage along x-axis. The graph will be as shown in figure.

Graphs

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Precautions

(i) Use voltmeter and milliammeter of suitable range.

(ii) Connect the Zener diode p-n junction in reverse bias.

(iii) The key should be kept open when the circuit is not in use.

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IDENTIFICATION OF DEVICES

EXPERIMENT # 3

Aim

To identify a diode, a L.E.D., a resistor and a capacitor from a mixed collection of such items

Apparatus

A multimeter and a collection of a junction diode, L.E.D., a resistor and a capacitor.

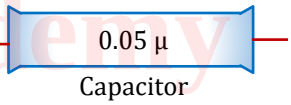
Theory

For identification of different items, we have to consider both, their physical appearance and working:

- A resistor, a capacitor, a diode and an LED are two terminal devices. For identifying these, we use the following facts :
  - (i) A diode is a two terminal device that conducts only when it is forward biased.
  - (ii) An LED is a light emitting diode. It is also a two terminal device which conducts and emits light only when it is forward biased.
  - (iii) A Resistor is a two-terminal device. It conducts both with D.C. and A.C. voltage. Further, a resistor conducts equally even when terminals of D.C. battery are reversed.
  - (iv) A capacitor is a two terminal device which does not conduct with D.C. voltage applies either way. But, conducts with A.C. voltage.

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Resistor



Capacitor


Diode


L.E.D.

Procedure

1. The component having two legs may either be a junction or capacitor or resistor or a light emitting diode. These items can be distinguished from each other by using a multimeter as an ohmmeter.
2. Touch the probes to the two ends of each item and observe the deflection on the resistance scale. After this, interchange the two probes and again observe the deflection.
3. (i) If the same constant deflection is observed in the two cases (before and after interchanging the probes), the item under observation is a resistor.
   (ii) If unequal deflections are observation, it is a junction diode.
   (iii) If unequal deflections are observed in the two cases along with emission of light in the case when deflection is large, the item under observation is an LED.
   (iv) On touching the probes, if a large deflection is observed, which then gradually decreases to zero the item under observation is a capacitor.

In case the capacity of the capacitor is of the order of picofarad, then the deflection will become zero within no time.

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Result

When the item is observed physically

S.No.	Number of legs (or pins) of the item	Inference
1	Two	Junction diode, L.E.D., resistor or capacitor

With multimeter as an ohmmeter:

S.No.	Possible deflection before and after interchanging the probes	Inference
1	Same constant deflection	The item is a resistor
2	Small deflection in one case and large deflection in the other.	The item is a junction diode
3	Small deflection in one case and large deflection in the other along with emission of light	The item is an L.E.D.
4	Large deflection, which gradually falls to zero	The item is a capacitor of small capacity

Precautions

Observe all those precautions which were related to multimeter and explained at the end of multimeter.

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